Resolution Geometry

Real Measurements:
Calibrate Once, Compute Many

A plug-and-play support paper for the print edition, on real GPS observations

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Abstract

A support paper for the print edition [1] must show that a framework's calibration is not self-serving. Fitting a model and then checking it on the data it was fitted to proves nothing. We run the physical-distance strata of Resolution Geometry (RG) on *real* measurements: dual-frequency GPS observations from two GEONET stations of the Geospatial Information Authority of Japan, 3.3 km apart, recorded on 2 April 2005 and processed with the broadcast ephemeris. The published station coordinates are external anchors.

The design separates calibration from use.

- One calibration is made, on station A only, from A's first half hour and A's anchor.
- It is frozen and then used for computations it never saw: station B's position, the A–B baseline, B's uncertainty coverage, and the swap in which B calibrates and A predicts.
- Independent observables are checked: the ionosphere is measured with carrier phase, which the positioning never uses.
- Every RG layer is removed in turn and must visibly fail.

Results:

- uncalibrated single-point positions agree with the anchors to 3.4 m (rms);
- with A's frozen calibration, B's error halves to 1.6 m, and the baseline agrees with the published one to 0.30 m;
- the calibrated uncertainty predicts B's per-axis coverage (0.947) and the swap (0.942) at the 0.95 target;
- phase-derived ionospheric variations (0.45 m) agree between stations to 0.009 m;
- removing the Sagnac term degrades positions to 27.6 m, the relativistic clock term to 8.7 m, frequency diversity to 5.9 m, and the clock nuisance to more than 10^5 m.

An access log certifies that station B's anchor is never read by an estimator. All ten checks pass, and two canaries are caught. Several pre-registered expectations failed on contact with the data. They are reported with their corrections (Section 6). The companion synthetic suite (B01–B14) is retained as an implementation-regression layer, and now runs under NumPy 2.

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1 What counts as support

Three layers of evidence are kept apart.

1. *Implementation regression* (B01–B14, synthetic). Data are simulated from the model that decodes them. This shows that the code implements the rungs and that its negative controls bite. It is *not* evidence about the world.

2. *Held-out real measurements* (RD01–RD10). These are data the framework did not generate, external anchors, a single frozen calibration, and computations it never saw.

3. *Ablations* (canaries on real data). Each RG layer is removed in turn and the result must degrade by more than its uncertainty.

Only layer 2 and layer 3 support RG empirically. A result that passes layer 1 but fails layer 2 is a correct implementation of a wrong model.

2 The data

stations	GEONET 0759 (A) and 3040 (B), Kanto, Japan; Trimble 5700 receivers
session	2005-04-02, 00:00–01:00 GPST, 30 s, 120 epochs each, 9–11 satellites
observables	C1 (C/A code), P2 (P code on L2), L1 and L2 carrier phase
orbits and clocks	GPS broadcast ephemeris (RINEX 2.10), 162 records
anchors	published station coordinates (RINEX header); baseline 3335.425 m
source	public RTKLIB test-data set [5] (`test/data/rinex`), GEONET [6]; SHA-256 recorded

The anchors are used only for evaluation, with one exception: station A's anchor in the declared calibration step. The access log (RD10) enforces this.

Why GPS. GPS is the most heavily tested real instance of the physical-distance strata:

- a counted timing channel with exact SI constants;
- a receiver clock that must be treated as nuisance;
- a stationary but non-static slice (the rotating Earth);
- a dispersive and a non-dispersive medium;
- external anchors.

The same pipeline accepts any RINEX observation and navigation pair, including the IGS archives that NASA distributes through CDDIS (Section 8). NASA servers were not reachable from the environment in which this paper was produced, so no NASA-hosted file was processed here.

3 The RG pipeline, step by step

Each step is one operation of the RG calculus (README, "The calculus"), with the rung it instantiates.

RG step	GPS realization	rungs
declare	protocols C1, P2 (code) and L1, L2 (phase); task: station position; media models declared	L0, H-med
counted channel	pseudorange = $c\times$ counted code delay, with c exact (SI)	L94–L96
anchor	satellite positions in metres (broadcast); external station coordinates held out	L99–L101, L109
nuisance	receiver clock estimated per epoch; efficient (Schur) covariance	L34
stationary slice	light-time with Earth rotation (Sagnac correction)	L104–L105
duration	satellite clock polynomial plus relativistic eccentricity term	L81–L82, L93
dispersive medium	ionosphere removed by frequency diversity (iono-free combination)	L107
non-dispersive medium	troposphere: Saastamoinen, standard atmosphere (declared)	L107, H-nuis
calibration	once, on A; frozen; transferred	L16, L114
certify	per-axis coverage; window of validity	L16, L47–L48, L114

The orbit, clock and relativistic formulas follow the public GPS interface specification [2, 4], and the troposphere model is Saastamoinen's [3]. Nothing is tuned to these stations.

4 Results

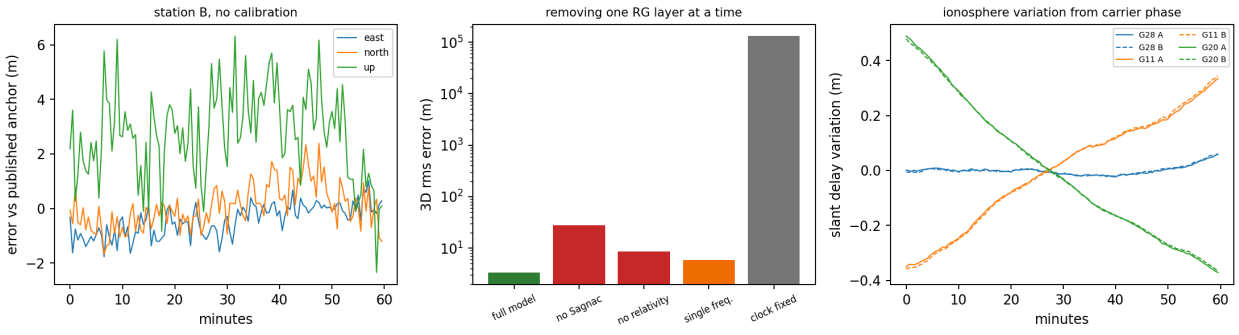


Figure 1: Real data. Left: station B errors against its published anchor, with no calibration. Centre: 3D rms error when one RG layer is removed. Right: ionospheric variation from carrier phase for three satellites; solid lines are station A, dashed lines station B.

id	check	result
RD01	provenance	120/120 epochs; 11 satellites; hashes recorded
RD02	no calibration: positions vs published anchors	rms 3D 3.51 m (A), 3.36 m (B); horizontal 1.15 and 1.08 m; mean up bias +2.85 and +2.73 m

id	check	result
RD03	calibrate the uncertainty once on A (white plus common floor); predict elsewhere	per-axis coverage: B from A 0.947; A from B 0.942; A's own second half 0.889 (time transfer weaker); white-only model fails
RD04	calibrate differential corrections once on A; frozen; apply to B	B's later half: 1.59 m (raw 3.36 m); baseline 3335.717 m against 3335.425 m published, error 0.30 m
RD05	remove Sagnac (synchronization holonomy)	27.5 and 27.6 m
RD06	remove relativistic clock term	8.73 and 8.69 m
RD07	frequency diversity; absolute and phase-derived medium	single-frequency 6.18 and 5.87 m; absolute code delay negative in 99% of samples (instrument biases); phase variations 0.452 m agree to 0.009 m between stations
RD08	zenith delay on A's anchor vs declared atmosphere	3.07 m estimated against 2.37 m model; correlation with height -0.93 when position is free
RD09	clock nuisance	clock fixed: 1.3×10^5 m (A), diverges (B); receiver offset 676 km; Schur covariance error 8×10^{-16}
RD10	non-circularity guard	B's anchor read only for evaluation; peeking pipeline flagged

4.1 Reading the results as RG statements

Counted channel and anchors (L94–L101, L109). Distances in metres arise only because time counts are multiplied by the exact SI c and referred to satellites whose positions are in metres. With these anchors, real positions agree with independently published coordinates at the few-metre level (RD02). Nothing about the stations was fitted.

Nuisance (L34). The receiver clock offset is 676 km in range units. Treating it as nuisance is what makes positioning possible at all (RD09). The efficient covariance is the Schur complement, to machine precision.

Stationary, non-static slice (L104–L105). The Earth-rotation correction is the synchronization holonomy of a rotating frame. Omitting it costs 24 m (RD05). The non-static slice is the operative geometry of every GPS fix.

Media (L107) and a genuine fibre. Frequency diversity resolves the dispersive medium for positioning (RD07). The *absolute* geometry-free code delay, however, is negative in 99% of samples, which no physical delay can be. Instrument inter-frequency biases share the fibre of the medium, so the absolute slant delay is a CORRESPONDENCE until those biases are declared. The *variations* are resolved, by an independent observable: carrier phase, which the positioning never uses. They agree between stations to 1 cm.

Non-dispersive medium and nuisance coupling (RD08). With A's anchor held, the zenith delay estimated from elevation diversity is within 0.7 m of the declared standard atmosphere. With position free, it correlates with height at -0.93, a near nuisance-rank wall. The troposphere must be declared, or the anchor held, which is exactly what H-nuis states. The $+2.8$ m up bias of RD02 points the same way: vertical errors absorb the medium.

5 Non-circularity

1. *One calibration.* Calibration uses station A only: its first half hour and its anchor.
2. *Frozen.* No parameter is refitted afterwards.
3. *Many uses.* The frozen calibration serves four computations it never saw:
   - B's position, in B's later half;
   - the A–B baseline;
   - B's uncertainty coverage;
   - the swap, with B calibrating and A predicting.
4. *Independent observables.* The medium is checked with carrier phase, which positioning does not use. The zenith delay is checked against a declared model it was not fitted to.
5. *Guarded anchors.* Every anchor access is logged with its purpose. B's anchor is used only for evaluation, and a peeking pipeline is caught (RD10).
6. *Ablations.* Each removed layer degrades the held-out result by far more than its uncertainty (RD05–RD07, RD09).

Remark 5.1 (What transfers, and what does not). The calibration transfers across *space*: B from A and A from B reach 0.947 and 0.942 per-axis coverage. It transfers less well across *time*: A's own second half reaches 0.889. Real errors drift within the hour, and RG reports such a calibration with its window rather than as a global lock (L114).

6 Registration record

These expectations were written before the real data were run and then failed. Each is corrected in the open; none was hidden.

1. *Coverage gate* (RD03). The first gate required all three axes inside their intervals at once, whose nominal rate is $0.95^3 = 0.857$, and set the bar at 0.85. It was re-registered as per-axis coverage, nominal 0.95. Joint coverage is still reported: 0.842 in both spatial transfers.
2. *Aging* (RD04). Frozen corrections were expected to age within the hour. They did not, and the claim became a validity window of at least 30 min.
3. *Instantaneous corrections* (RD04). These were expected to be best. With epochs matched correctly they are noise-dominated (2.27 m), because they copy A's amplified code noise into B. The averaged, frozen correction is the meaningful one.
4. *Sign of the medium* (RD07). Positive absolute slant delays were expected. They are negative, which revealed the instrument-bias fibre. A code-based variation test was then tried and was swamped by code noise (0.86 m); the test was moved to carrier phase.
5. *Bugs.*
   - The two receivers tag epochs milliseconds apart, so exact-time keys matched almost nothing. Epochs are now matched on the nominal grid.
   - With the clock fixed to zero, the solver diverged into an exception instead of reporting a destroyed solution.

No threshold was relaxed to rescue a claim without the claim being changed to what the data support.

7 The synthetic regression layer

The companion suite B01–B14 covers fourteen cases:

- calibrated radar, correlated triangulation, clock-nuisance multilateration and anisotropic anchors;
- sphere holonomy, curved radar against rods, the Sagnac ring and FLRW radar;
- fold correspondences, medium scale ambiguity and cone reconstruction up to scale;
- Fisher length against metres, a single anchor against held-out validation, and systematic floors.

It passes 14/14 on five seeds with 14/14 negative controls. One portability fix was needed: NumPy 2 changed the semantics of batched `solve`, and two calls now carry an explicit trailing axis. The suite tests that the rungs are implemented correctly, and it is cited as such, not as evidence.

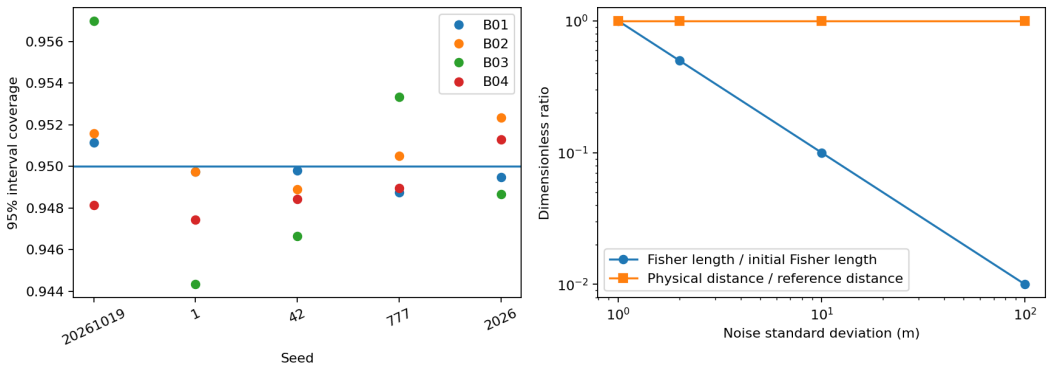


Figure 2: Synthetic regression layer (B01–B04 coverage across seeds; Fisher length against anchored distance).

8 Plug and play

1. *Input.* Supply a RINEX 2 observation file for each station (C1, P2, L1, L2) and one navigation file. Any IGS station from CDDIS or another archive works, as do two stations with published coordinates.
2. *Declare.* Choose the calibration station and window, the media models, and the elevation mask.
3. *Run.* `python3 rg_gnss_real_attack.py --data DIR --figs figs`.
4. *Read.* Each output line is an RG statement with its rung. The access log shows which anchors were read and why.
5. *Attack.* `--canary` must fail. Removing a layer must degrade the held-out result.

The code reads station names from the headers and assumes nothing else about the stations. Replacing the data replaces the experiment, while the RG statements and their gates stay fixed.

9 What this supports, and what it does not

Supported. On real measurements:

- the counted channel with SI anchors yields metres;

- clock nuisance, the Sagnac holonomy, the duration correction and frequency diversity each carry measurable, necessary weight;
- one calibration transfers to independent computations;
- the medium is resolved only up to an instrument-bias fibre, exactly as RG's media rung predicts.

Not supported, or not claimed.

- This is one hour and two stations, not a survey.
- Broadcast rather than precise orbits limit the accuracy to the metre level.
- The header coordinates are taken as published anchors; their own uncertainty is not re-derived.
- No NASA mission data were processed here.
- The results support the physical-distance strata; they do not test the quantum, residual-characteristic or chaos strata.
- The data are deterministic (no seeds), so repetition across sessions and stations, not seeds, is the next step.

A Reproducibility

- `rg_gnss_real_attack.py` (RD01–RD10) passes 10/10. Its canaries are a white-only uncertainty model and a pipeline that peeks at B's anchor; both are caught.
- `rg_geometry_benchmark.py` (B01–B14) passes 14/14 on five seeds.
- The data files (RINEX) and their SHA-256 hashes are bundled.

References

[1] M. Rada, *Resolution Geometry* (print edition), 2026.

[2] IS-GPS-200, *Navstar GPS Space Segment / Navigation User Interfaces*, GPS Directorate.

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[4] N. Ashby, Relativity in the Global Positioning System, *Living Rev. Relativ.* 6 (2003), 1.

[5] T. Takasu, RTKLIB: an open-source program package for GNSS positioning (test data, `test/data/rinex`).

[6] Geospatial Information Authority of Japan, GEONET GNSS continuous observation system.